NATHANIEL (NATE) K. MINA

CEO & MECHANICAL ENGINEER | MARINE IT & FIBER INFRASTRUCTURE | SYSTEMS ARCHITECT

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EXECUTIVE SUMMARY & SYSTEMS PHILOSOPHY

Mechanical Engineer, Marine IT Network infrastructure and fiber networks (RIT B.S.) & Systems Architect with 10+ years of expertise in custom computer builds, IT infrastructure, marine network systems, fiber optic networks, bespoke liquid-cooled hardware, and Microsoft Power Automate / Generative AI workflows. Seasoned Mechanical Engineer, Marine IT Network infrastructure and fiber networks with over 10 years of experience in custom computer builds, IT systems architecture, marine network infrastructure, fiber optics, and high-performance computing hardware. Grounded in my mechanical engineering education at Rochester Institute of Technology (RIT), I approach technology through a first-principles lens. I treat digital infrastructure, liquid thermal management, local AI integration, and process automation as coupled, deterministic systems that can be modeled and optimized for peak efficiency.

CORE COMPETENCIES & TECHNICAL STACK

- **Mechanical Engineering & Design:** SolidWorks, AutoCAD, Finite Element Analysis (FEA), Strength Testing, Rotating Machinery, Liquid Cooling Thermal Design
- **Languages & Automation:** Python, TypeScript, HTML/CSS, Microsoft Power Automate
- **AI Ecosystems & Computing:** Generative AI, OpenAI API, GitHub Copilot, Prompt Engineering, Multi-GPU Compute Nodes
- **Hardware & Infrastructure:** Marine Network Infrastructure, Bespoke Workstation Design, Data Forensics & Recovery, Network Administration
- **Methodologies & Leadership:** 5-Whys Root Cause Analysis, Agile Project Management, Process Mapping, Cross-Functional Leadership

PROFESSIONAL EXPERIENCE

Chief Executive Officer & Founder
January 2008 - Present

Pure Computers | Rochester & Henrietta, NY

- Lead business operations and IT architecture at Pure Computers (www.purecomp.net), providing custom high-performance computer builds, component repairs, and managed IT services.
- Design, assemble, and optimize bespoke liquid-cooled workstations, high-throughput gaming PCs, and multi-GPU compute nodes tailored for demanding workloads.
- Deliver full-service IT infrastructure solutions, network administration, system upgrades, hardware diagnostics, and data recovery for business and individual clients.
- Implement Microsoft Power Automate, local AI integration, and automated customer ticketing pipelines to enhance operational efficiency.
- Provide expert hardware troubleshooting, component-level repair, and ongoing IT support with a commitment to zero downtime.

Network Engineer & Marine Networking Project Administrator
2010 - 2017

Waveguide Communications | Pompano Beach, FL & Germany

- Designed and developed Marine I.T. Network Infrastructure for maritime vessels globally.
- Conducted Finite Element Analysis (FEA) to optimize structural cable routing and physical enclosure durability in harsh marine environments.
- Managed large teams of technicians and cable pullers, ensuring strict adherence to maritime deadlines and specifications.

Project Designer & Technical Design Lead
Munich, Germany

Siemens Energy (Siemens Energy Global GmbH & Co. KG) | Munich, Germany

- Served as project technical design lead for rotating machinery engineering projects.
- Provided technical leadership and support to a multidisciplinary engineering team throughout the project lifecycle from sales handover to execution.
- Led a team of 10 engineers in developing and implementing new software/hardware products, launching three products generating $10 Million in revenue with a $1M budget.

Customer Service Representative
October 2018 - October 2021

ADT LLC | Henrietta, NY

- Managed customer de-escalation, billing, technical troubleshooting, alarm verification, and emergency dispatch response.
- Served as team lead, developing streamlined call center interaction protocols and customer service workflows.

Impact Team Member
2009 - 2011

Hollister Co. | Henrietta, NY

- Managed inventory logistics, merchandise processing, and visual asset arrangement.

EDUCATION & ACADEMIC BACKGROUND

Bachelor of Science (B.S.) in Mechanical Engineering | Minor in Mathematics & Computer Science Sept 2007 - Aug 2012

Rochester Institute of Technology (RIT) — Rochester, NY

- Primary field of study focusing on Solid Mechanics, Thermal Systems, CAD, and FEA Analysis.
- Completed minor in Mathematics and coursework in Computer Science.
- Dean's List Honoree (2007, 2009, 2010).

Advanced Regents High School Diploma Sept 2003 - June 2007

Rush-Henrietta Senior High School — Henrietta, NY

- Advanced Regents Diploma with honors in mathematics and physical sciences.

CERTIFICATIONS & CREDENTIALS

- **Introduction to Generative AI** – Google / Coursera
- **Search Engine Optimization (SEO) with Squarespace** Coursera Project Network
- **Conduct a 5 Whys Analysis in Miro** – Coursera Project Network
- **Use Canva to Create Desktop and Mobile-Friendly Web Pages** Coursera Project Network

PROVEN METRICS & SYSTEMS CASE STUDIES

Pull House Heat to Pool (ThermalCore v5.0) (Thermal & Systems Engineering) **+34.2% Efficiency [COP Gain]**

Integrated sub-ambient liquid heat exchange and atmospheric evaporation system for total home-pool thermal equilibrium, replacing gas heaters with waste AC heat recovery. Impact: Recovers up to 48,000 BTU/hr of waste AC heat to pool water, cutting electric cooling load by 32% and displacing $340+/mo in natural gas.

VitalStats Clinical Supplement Matrix (Analytics & Clinical Portal) **97.0% Max Synergy [SUCRA Score]**

Integrated CDC NHANES surveillance datasets with RCT meta-analyses to engineer an interactive diagnostic assessment wizard and compound evidence matrix. Impact: Synthesized 15+ clinical compounds with automated contraindication detection and age/sex disaggregated drivers.

Manufacturing Process Optimization (Industrial Systems) **+15% [Output Gain]**

Applied a formal 5 Whys root-cause analysis and rigorous process mapping to isolate mechanical and logistical bottlenecks for a regional manufacturing partner. Impact: Eliminated 18 weekly hours of line stoppage and stabilized throughput variance.

Custom Deep Learning Workstation (Hardware Engineering) **3.0x Baseline [Simulation Speed]**

Designed and fabricated a liquid-cooled, multi-GPU compute node for a university research team, optimizing thermal dissipation paths to prevent throttling under continuous loads. Impact: Reduced tensor training iterations from 72 hours down to 24 hours under peak thermal equilibrium.

AI-Driven Customer Ingest Pipeline (Workflow Automation) **-40% [Latency Drop]**

Deployed an end-to-end automated ingest pipeline utilizing intelligent classification models coupled to custom backend hooks. Impact: Automated 85% of incoming ticket categorizations while maintaining zero data degradation.